



NIH Websites Are Changing

NIH is moving the launch of its new website to the fall to ensure the best possible experience for everyone who relies on our websites for health information, research, funding opportunities, and other resources.

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Preventing Diabetes

Small Steps Can Make a Big Difference

Chances are, you know someone who has diabetes. It might even be you. Diabetes is one of the most common disorders in the U.S. It affects about 1 in 9 Americans. Diabetes raises your risk for serious health problems. It can damage the eyes, kidneys, nerves, and heart, and it is linked to some types of cancer.

 Illustration of a group Zumba class.

Now, what if you learned that there's a low-cost, scientifically proven way to greatly reduce your chances of getting type 2 diabetes, the most common type? Would you give it a try?

More than two decades ago, a landmark NIH-supported study, called the Diabetes Prevention Program (DPP), released its results. It found that people at high risk for diabetes were much less likely to get the disorder if they lost a little weight

Wise Choices

Aim to Prevent Diabetes

- **Maintain a healthy weight.** If you are overweight, set a weight-loss goal. Start by aiming to lose at least 5% of your current weight.
- **Get moving.** Get at least 30 minutes of physical activity 5 days a week.
- **Eat healthy.** Find tips for healthy eating at www.myplate.gov.
- **Get support.** [CDC's National Diabetes Prevention Program \(NDPP\)](#) offers year-long programs nationwide. Participants work with a lifestyle coach in person or online to build healthy habits that last. It's free to eligible participants. Find an NDPP near you.

through healthy eating and physical activity. Their risk of getting diabetes dropped by nearly 60% compared to people who did not aim to make healthy changes.

For those over age 60, the results were even more striking. Their risk of getting diabetes dropped by 71% when they made healthy changes. The benefits were so clear that the study ended a year early, after just three years. Participants in the comparison group were encouraged to also make the healthy changes to reduce their diabetes risk.

DPP has had a lasting influence on medical care in the U.S. and around the world. Since its initial results were reported, lifestyle change programs based on the findings have become widely available across the country.

A 10-year follow-up study showed that people in the original treatment group delayed diabetes by about four years. A later study found benefits even 22 years after the study began, with people in the lifestyle-change group having about a 25% reduced risk of developing diabetes.

“Even though the initial treatments lasted only three years, participants did have longer-term health benefits,” says Dr. William Knowler, an NIH diabetes expert.

NIH-supported researchers continue to study new and proven ways to help people prevent or delay type 2 diabetes. But we already know that taking steps to prevent or manage diabetes can lower your risk of developing diabetes-related health problems.

Are You at Risk?

Diabetes is a disease that occurs when your blood glucose, also called blood sugar, is too high. Glucose is your body’s main source of energy. Normally, a **hormone** made by the pancreas called insulin helps glucose get into your cells to be used for energy. If you have diabetes, your body doesn’t make enough insulin or use insulin properly. Glucose then stays in your blood and doesn’t reach your cells.

Anyone can get type 2 diabetes, even children. But certain factors can raise your risk. You’re more likely to develop type 2 diabetes if you are at least 35 years old or have a family history of diabetes. Black Americans, Hispanic/Latino Americans, American Indians, and Asian Americans are also at higher risk.

Find More Information

- [Managing Diabetes: New Technologies Can Make It Easier](#)
- [Preventing Type 2 Diabetes](#)
- [Diabetes Overview](#)
- [Diabetes Prevention Program](#)
- [National Diabetes Prevention Program \(CDC\)](#)
- [Medicare Diabetes Prevention Program](#)

References

[Effects of metformin and intensive lifestyle interventions on the incidence of kidney disease in adults in the DPP/DPPOS.](#)

Molitch ME, Tripputi M, Levey AS, Crandall JP, Dabelea D, Herman WH, Knowler WC, Orchard TJ, Schroeder EB, Srikanthan P, Temprosa M, White NH, Nathan DM; Diabetes Prevention Program Research Group. *J Diabetes Complications*. 2023 Sep;37(9):108556. doi: 10.1016/j.jdiacomp.2023.108556. Epub 2023 Jul 25. PMID:37607422.

[Effects of long-term metformin and lifestyle interventions on cardiovascular events in the Diabetes Prevention Program and its outcome study.](#)

Goldberg RB, Orchard TJ, Crandall JP, Boyko EJ, Budoff M, Dabelea D, Gadde KM, Knowler WC, Lee CG, Nathan DM, Watson K, Temprosa M; Diabetes Prevention Program Research Group.* *Circulation*. 2022 May 31;145(22):1632-1641. doi: 10.1161/CIRCULATIONAHA.121.056756. Epub 2022 May 23. PMID:35603600.

[10-year follow-up of diabetes incidence and weight loss in the Diabetes Prevention Program Outcomes Study.](#)

Diabetes Prevention Program Research Group. *Lancet*. 2009 Nov 14; 374(9702): 1677–1686. PMID: 19878986.

People who are overweight or have obesity and people who don't get enough physical activity are also more likely to develop the disease. But these are changeable factors.

About 1 in 3 American adults has a condition called prediabetes. It occurs when your blood sugar is higher than normal, but not yet high enough to be called diabetes. Most people with prediabetes don't realize they have it. That's because prediabetes tends to have few symptoms, and many people don't get screened.

"We can diagnose prediabetes and diabetes with a very simple blood test known as a hemoglobin A1C. You don't need to fast to do this test," says Dr. Tannaz Moin, a physician and diabetes researcher at the University of California, Los Angeles. The A1C test reflects your average levels of blood sugar over the past three months.

"Prediabetes is viewed as a very strong risk factor for type 2 diabetes. But not everyone with prediabetes goes on to develop type 2 diabetes," Moin adds. "That's why it's important to be screened and talk with your medical providers about your risk factors and things that you can do to prevent type 2 diabetes if you're at risk."

A Role for Medications

Although making healthy lifestyle changes has proven effective for preventing diabetes, it doesn't work for everyone. For those people, medications may help.

NIH's DPP study looked at whether the diabetes drug metformin might also prevent or delay diabetes onset. It found that the drug could reduce the risk of developing diabetes by about 30%. For some patients, a combination of metformin and lifestyle changes might be best.

In recent years, a class of drugs called GLP-1 drugs has become widely available for weight management and diabetes treatment. These drugs have proven effective at reducing weight and keeping blood glucose in check. Some studies hint that they might also help to prevent or delay diabetes.

"Body weight is an important factor that determines the risk of getting type 2 diabetes. So the potential here is great," says Knowler. "But the problem is, as with any new drugs, it takes many years to know how effective and safe they will be. And a huge problem with any kind of weight loss is sustaining it, not achieving it."

[Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin](#). Knowler WC, Barrett-Connor E, Fowler SE, Hamman RF, Lachin JM, Walker EA, Nathan DM; Diabetes Prevention Program Research Group. *N Engl J Med*. 2002 Feb 7;346(6):393-403. doi: 10.1056/NEJMoa012512. PMID:11832527.

Because GLP-1 drugs are still relatively new, it's not entirely clear how outcomes will change if people stop taking the drugs. And possible side effects of long-term use are still uncertain.

Healthy Changes That Last

"Making lifestyle changes is hard. So it's important to get the support you need to make the changes last," says Dr. Joshua J. Joseph, a physician and diabetes researcher at Ohio State University. "You can get that support through programs like the [CDC's National Diabetes Prevention Program \(NDPP\)](#)."

NDPP is based on the findings of NIH's DPP study. It aims to help people make long-term lifestyle changes that prevent or delay diabetes (see the Wise Choices box).

Local hospitals, health departments, libraries, senior centers, and faith-based organizations may also offer programs or seminars to help prevent type 2 diabetes. For physical activity, look for activities that you enjoy. Experts recommend brisk walking for a low-cost activity. Working out with others can be motivating for some people. Consider group classes like Zumba or Pilates.

"It's important to think of diabetes prevention throughout the lifespan. So, it's just as critical to work with children on healthy eating and healthy behaviors as it is to work with older people," Joseph explains. "If we work across the life course, it could lead to large reductions in the development of diabetes in the U.S. and around the world."

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